

COMP0051 Algorithmic Trading Coursework

1 Time Series Prep

1.1 Introduction and Data Sources

Daily closing price data for the SPDR Portfolio Long Term Treasury ETF (SPTL) was gathered using yfinance [7], covering the period from 1 January 2023 to 31 December 2023. The Effective Federal Funds Rate (EFFR Index), used as the risk-free rate, was downloaded from the Federal Reserve Bank of New York website [3]. The two datasets were merged into a single dataframe, with missing dates forward-filled using the previous day's values.

1.2 Risk-Free Rate Conversion

The EFFR data is annualized and in order to convert it into a daily rate, we had to divide it by 252 (the approximate number of trading days in a year). For each day t ,

$$r_t^f = EFFR(t) * \frac{1}{252}$$

This daily risk-free rate of return represents the theoretical return an investor would make on an investment with zero risk. [5] It can be used as a benchmark against the returns from other investments.

1.3 Daily Excess Return Calculation

The daily excess return of holding SPTL is defined as:

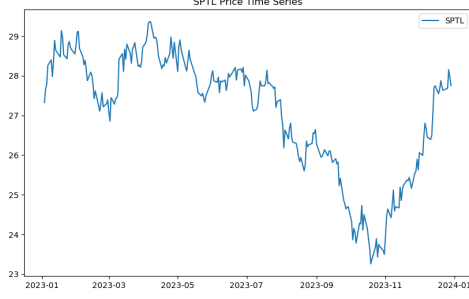
$$r_t^e = \frac{\Delta p_t}{p_{t-1}} - r_{t-1}^f = \frac{p_t - p_{t-1}}{p_{t-1}} - r_{t-1}^f$$

where p_t is the closing price of SPTL at time t and r_f^t is the daily risk-free rate. This captures the return from investing in SPTL in excess of the risk-free rate.

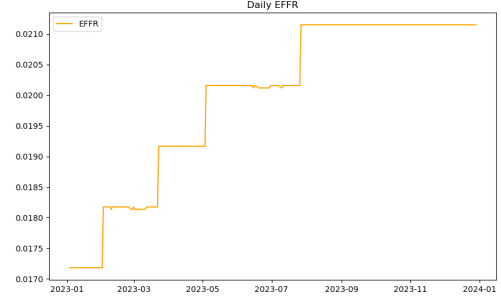
1.4 Visualizing SPTL, EFFR, and Excess Returns

Observations from the figures below:

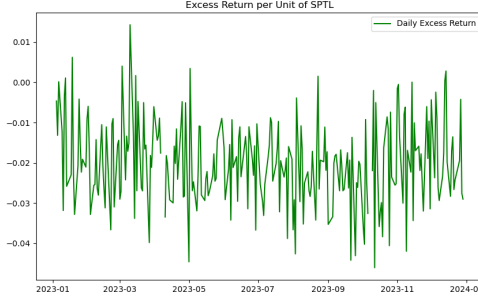
- The SPTL etf showcases a steady first half of the year (with a slightly negative trend). It then has a largely declining trend for the next quarter of the year. In the final quarter, the trend reverses upward, and the ETF ends the year approximately flat relative to its starting value.
- The EFFR rate increases in steady increments throughout the year. This reflects policy adjustments from the Fed.
- The daily excess returns fluctuate around -2% and even occasionally dipped below -4%. It had a few positive days, with only one being above 1%. This shows that over the year, the EFFR daily rate was typically higher than that of the SPTL daily percentage change.



(a) Closing price of the SPTL etf throughout 2023



(b) EFFR Daily Rate in 2023



(c) Excess Returns per Unit of SPTL in 2023

2 Trading Strategies

2.1 Leveraged Strategies

Two leveraged trading strategies were employed on SPTL. Both started with an initial capital of $V_0 = \$100,000$ and a leverage of $L = 10$. The leverage implies that at any time t , the dollar value of the position must be less than or equal to the total capital value times the leverage $|\theta_t| \leq V_t^{total} * L$. The uninvested portion of capital, given by $V_t^{total} - \frac{|\theta_t|}{L}$, earns the daily risk-free rate r_t^f .

2.2 Trading PnL

For both strategies, we calculate the profit and loss ΔV_t^{total} for each day which comprises two components.

1. Trading PnL ΔV_t

$$\left(\frac{\Delta p_t}{p_{t-1}} - r_{t-1}^f \right) \theta_{t-1} = \left(\frac{p_t - p_{t-1}}{p_{t-1}} - r_{t-1}^f \right) \theta_{t-1},$$

where:

- p_t is the SPTL closing price at time t
- θ_t is the dollar value of SPTL at time t (i.e. $\theta_t = \text{units}_t \times p_t$)
- r_t^f is the daily risk-free rate

2. Capital Account Growth ΔV_t^{cap}

The return on the portion of capital not used as margin for the SPTL position.

$$\Delta V_t^{cap} = (V_{t-1}^{total} - M_{t-1}) r_{t-1}^f$$

where:

- V_t^{total} is the total capital value

- M_t is the total margin used with $M_t = \frac{|\theta_t|}{L}$
- r_t^f is the daily risk-free rate

The total daily change in capital is:

$$V_t^{total} - V_{t-1}^{total} = \Delta V_t^{total} = \Delta V_t + \Delta V_t^{cap}.$$

We track V_t over time by accumulating these daily changes starting with $V_0 = \$100,000$.

2.3 Moving Average Convergence-Divergence (MACD) Momentum Strategy

1. Trading Signals

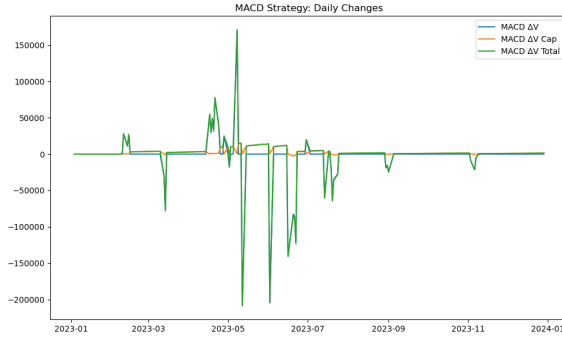
- The MACD strategy uses the difference between the 26-day and 12-day exponential moving averages (EMA) of the SPTL price to generate momentum signals.[4]
- In this momentum strategy, a positive MACD indicates a bullish signal while a negative MACD indicates a bearish signal.

2. Positioning Rules

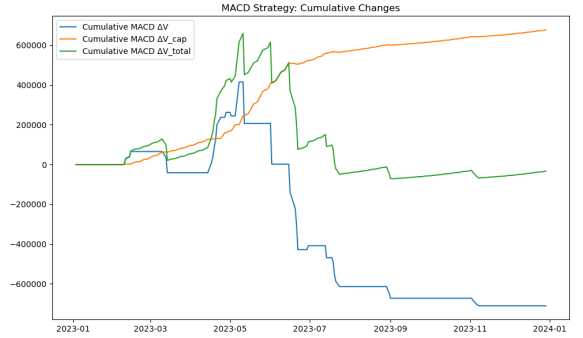
This strategy is based on the MACD and the price of SPTL in relation to the 26 day EMA.

- A long position is taken if the MACD is positive and SPTL price is above the 26 day EMA
- A short position is taken if the MACD is negative and SPTL price is below the 26 day EMA
- If these conditions are not met, we hold no SPTL (i.e. $\theta_t = 0$)

3. Results and Figures



(a) MACD Strategy: Daily Changes



(b) MACD Strategy: Cumulative Changes

Daily Changes

- What it shows: The ΔV (blue) shows the day to day changes and this exhibits large spikes when the strategy significantly changes its position. ΔV_{cap} (orange) is more stable which reflects the smaller, steadier returns from the uninvested capital. ΔV_{total} (green) is the sum of both which occasionally swings sharply positive or negative on signal flips.
- Key Insights: In late Q1 and early Q2, we see noticeable spikes. These are both positive and negative which shows that the MACD signals triggered sizable position changes

Cumulative Changes

- What it shows: The cumulative ΔV (blue) starts off positive but then drops drastically mid-year. This signals a period of sustained losses in the trading component. The ΔV_{cap} (orange) grows at a steady rate as uninvested capital accrues the risk-free rate. The green line (cumulative total) reflects the sum of these two.

- Key Insights: We observe a deep drawdown in the trading portion, which pulls the total performance (green line) into negative territory in the latter half of the year. The amount invested in the risk free rate (orange) can only partially offset those trading losses.

2.4 Bollinger Bands Mean Reversion (BBMR) Strategy

1. Trading Signals

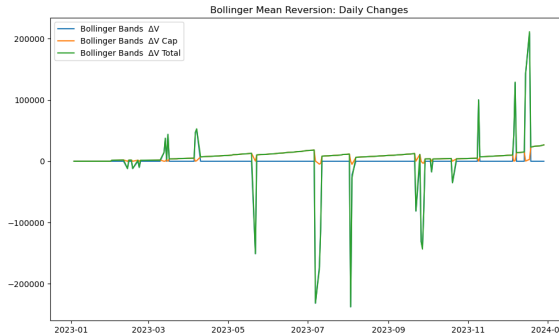
- The BBMR strategy assumes prices revert to their mean and uses Bollinger Bands to identify overbought/oversold conditions. This strategy creates an upper and lower bound of the mean. [1]
- This strategy uses the 20-day rolling mean (MA20) and its standard deviation (SD20) of the SPTL price.
- We calculate the upper band as 2 standard deviations away from the MA20 price (i.e. $UB = MA20 + 2 * SD20$). The lower band is calculated as 2 standard deviations below the moving average (i.e. $LB = MA20 - 2 * SD20$)

2. Positioning Rules

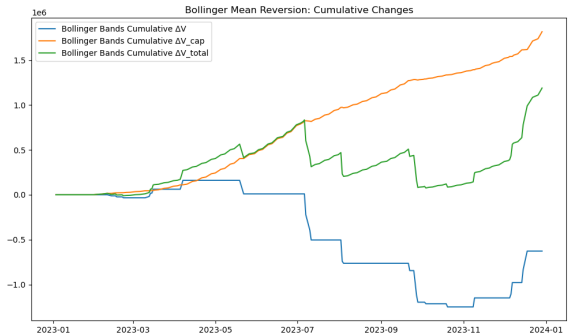
This trading strategy is based on the price being above or below the upper and lower bounds.

- A long position is entered if $p_t < LB_t$
- A short position is entered if $p_t > UB_t$
- If these conditions are not met, we hold no SPTL (i.e. $\theta_t = 0$)

3. Results and Figures



(a) Bollinger Mean Reversion Strategy: Daily Changes



(b) Bollinger Mean Reversion Strategy: Cumulative Changes

Daily Changes

- What it shows: The ΔV (blue) can produce large negative spikes if the strategy bets on a mean reversion just before a price swing. The capital portion ΔV_{cap} (orange) is once again smoother. The sum ΔV_{total} (green) occasionally jumps sharply positive, especially if prices revert quickly after overshooting.
- Key Insights: By design, mean reversion signals mainly accumulate profits when prices move back into the middle band. The negative spikes suggest periods when the ETF broke out further without immediate reversion. We see big negative spikes in the middle of the year, with large positive spikes towards the end of the year.

Cumulative Changes

- What it shows: The orange line (cumulative capital gains) grows steadily as uninvested funds earn the daily rate, while the blue line (cumulative trading PnL) sees periods of both profit

accumulation and significant downturns. The green line (cumulative total) shows the combined performance over time.

- Key Insights: Despite some big drawdowns in the trading portion in the middle of the year, the total ultimately trends upward, especially towards the end of the year. By year-end, the cumulative total remains well above zero.

3 Performance Indicators

3.1 Sharpe Ratio

The Sharpe Ratio (SR) is a measure of the risk-adjusted return. It does this by comparing the average of the returns above the risk-free rate to the volatility of the returns. [6] Formally this is calculated by:

$$SR = \frac{\mu}{\sigma} \sqrt{252},$$

where μ_{return} and σ_{return} are the mean and standard deviation, respectively, of the daily strategy returns $\left(\frac{\Delta V_{\text{total},t}}{V_{t-1}} \right)$.

A higher Sharpe ratio indicates a better performance relative to the risk taken. A Sharpe Ratio below 1 is generally considered poor; between 1 and 1.99 is adequate or good; 2 to 2.99 is very good; and above 3 is excellent [6].

3.2 Calmar Ratio

The Calmar Ratio (CR) is another measure of performance based on the amount of risk taken. It evaluates a strategy's returns relative to its maximum drawdown (the largest peak-to-trough decline). [2] It is defined as:

$$CR = \frac{\mu}{MDD} * 252$$

where

$$MDD = \frac{MinTroughValue - MaxPeakValue}{MaxPeakValue}$$

A higher Calmar suggests the strategy produced stronger returns for for each unit of maximum drawdown risk..

3.3 Comparison and Discussion

Strategy	Sharpe Ratio	Calmar Ratio
MACD	0.58	0.87
BBMR	2.84	4.12

- Over the year, the BBMR strategy performed considerably better and ended the year with \$1,289,119 while the MACD strategy ended the year with \$67,312. For the first third of the year, the performance was very comparable, and both had large drawdowns during the middle of the year. Only the BBMR strategy managed to fully recover—and even exceed—its mid-year losses in the final two months.
- By comparing the ratios by strategy, we can see that the BBMR strategy Sharpe Ratio significantly outperforms the MACD strategy. This implies that the BBMR Strategy had much higher returns per unit of volatility.
- Similarly for the Calmar Ratio, the BBMR strategy performed much better. This implies that the MACD strategy suffered deeper drawdowns without having returns that compensated for them.
- By looking at the overall performance and the two ratios, we can see that the BBMR strategy delivered better risk adjusted performance.

References

- [1] *Bollinger Bands Mean Reversion by Kevin Davey — Strategy by EdgeTools*. en. Oct. 2024. URL: <https://www.tradingview.com/script/um0CSa0t/> (visited on 03/24/2025).
- [2] *Calmar Ratio*. en-US. URL: <https://corporatefinanceinstitute.com/resources/career-map/sell-side/capital-markets/calmar-ratio/> (visited on 03/25/2025).
- [3] *Effective Federal Funds Rate - FEDERAL RESERVE BANK of NEW YORK*. URL: <https://www.newyorkfed.org/markets/reference-rates/effr> (visited on 03/23/2025).
- [4] *How to Trade the MACD*. en. URL: <https://www.investopedia.com/articles/forex/05/macddiverge.asp> (visited on 03/24/2025).
- [5] *What Is the Risk-Free Rate of Return, and Does It Really Exist?* en. URL: <https://www.investopedia.com/terms/r/risk-free-rate.asp> (visited on 03/23/2025).
- [6] *Sharpe Ratio*. en-US. URL: <https://corporatefinanceinstitute.com/resources/career-map/sell-side/risk-management/sharpe-ratio-definition-formula/> (visited on 03/25/2025).
- [7] Ran Aroussi. *yfinance documentation — yfinance*. URL: <https://yfinance-python.org/index.html> (visited on 03/23/2025).